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Impact of Photobiomodulation on T3/T4 Ratio and Quality of Life in Hashimoto Thyroiditis

Candas Ercetin, MD,¹ Nuri Alper Sahbaz, MD,² Sami Acar, MD,³
Firat Tatal, MD,⁴ and Yesim Erbil, MD⁵

Abstract

Objective and background: Hashimoto's thyroiditis (HT) is both a B cell- and T cell-mediated, organ-specific autoimmune disease. No current treatment for underlying pathological mechanisms is available for HT and once diagnosed it requires long-term levothyroxine (LT₄) treatment in most patients. The aim of our study was to evaluate the effects of photobiomodulation (PBM) on HT patients regarding thyroid functions, thyroid autoantibody levels, and decrease in hormone replacement needs.

Methods: A total of 350 patients, who were diagnosed with Hashimoto's thyroiditis, were included in our study. Patients were classified into two groups. Group 1 ($n=210$) received PBM and dietary supplementation such as vitamin D, iron, and selenium. Group 2 ($n=140$) received dietary supplements only. Patients' needs for LT₄ replacement levels, triiodothyronine (T3), thyroxine (T4), and TSH levels, T3/T4 ratio, and thyroid autoantibody levels were evaluated.

Results: As we compare both groups, the increase in T3 levels and T3/T4 ratio was markedly superior in Group 1 ($p=0.0001$). The decrease in thyroid peroxidase antibody (TPO Ab) levels was also significantly different between both groups ($p=0.0001$). Hormone replacement needs were also significantly decreased in Group 1 compared with Group 2 ($p=0.03$). Low-level laser therapy (LLLT) is 70 times more effective in increasing T3/T4 ratio ($p=0.001$) and 15 times more effective in decreasing levothyroxine dosage.

Conclusions: Our results are encouraging and PBM seems to be very effective in increasing T3/T4 ratio and decreasing TPO Ab levels and weekly dosages of LT₄ replacement therapy. Anti-inflammatory properties of PBM are greatly responsible for these changes and PBM causes major improvements in HT-related symptoms of the patient.

Keywords: photobiomodulation, Hashimoto thyroiditis, autoimmune thyroiditis

Introduction

HASHIMOTO'S THYROIDITIS (HT) is both a B cell- and T cell-mediated, organ-specific autoimmune disease. It is caused by an autoimmune attack of the thyroid gland and characterized by the presence of thyroid peroxidase antibodies (TPO Ab) and thyroglobulin antibodies.¹ Genetic and environmental factors were held responsible and it is the most common cause of hypothyroidism in noniodine-deficient populations.² The incidence and prevalence of HT are increased in the last decades and new therapeutic strategies need to be developed to effectively treat this public health issue.^{1,3}

No current treatment for underlying pathological mechanisms is available for HT and once diagnosed, it requires long-term levothyroxine (LT₄) treatment in most patients.⁴

Steroids or nonsteroid anti-inflammatory medication does not help in decreasing the destruction of thyroid gland. LT₄ replacement and selenium supplementation may reduce TPO Ab levels in some patients.⁵

In this setting, laser beam may be an alternative treatment option in HT. Light interacts with biological structures and improves anti-inflammatory responses and helps regeneration of tissues.⁴ Photobiomodulation (PBM) is shown to be effective in some autoimmune diseases, such as rheumatoid arthritis and Sjögren's syndrome.⁶ Also, PBM is shown to improve triiodothyronine (T3) and thyroxine (T4) levels in animal models.^{7,8}

PBM is a noninvasive, low-risk procedure. It does not contain any ionizing radiation. It is easy to perform and also low cost.⁶ The aim of our study was to evaluate the effects of PBM

¹Department of General Surgery, Bagcilar Training and Research Hospital, University of Health Sciences, Istanbul, Turkey.

²Department of General Surgery, Bakirkoy Dr. Sadi Konuk Training and Research Hospital, University of Health Sciences, Istanbul, Turkey.

³Department of General Surgery, Acibadem Taksim Hospital, Istanbul, Turkey.

⁴Department of General Surgery, Istanbul Kolan Hospital, Istanbul, Turkey.

⁵Department of General Surgery, Istanbul Medical Faculty, University of Istanbul, Istanbul, Turkey.

on HT patients regarding thyroid functions, thyroid autoantibody levels, and decrease in hormone replacement needs.

Materials and Methods

A total of 350 patients, who were diagnosed with Hashimoto's thyroiditis with the presence of thyroid autoantibodies and/or ultrasound findings of thyroiditis between January 2017 and June 2018, were included in our study. The female to male ratio was 340/10, and mean age of patients was 37 ± 8 . All patients were on hormone replacement therapy with LT_4 due to hypothyroidism and were in the normal range for serum T3, T4, and TSH concentrations. Patients with other autoimmune diseases, thyroid nodules, history of radiation to the neck, and previous thyroid surgery or other diseases, which can interfere with the thyroid metabolism, were excluded from the study.

Patients were classified into two groups. Group 1 ($n=210$) received PBM and dietary supplementation such as vitamin D, iron, and selenium. Group 2 ($n=140$) received dietary supplements only. All patients were included; none of them dropped out from the study. Turkey is endemic for vitamin D deficiency and patients with a level of <40 ng/dL received replacement, and also patients with iron levels <30 μ g/dL received iron replacement. A daily intake of 100 μ g of oral selenium is also given to all patients.

Patients' needs for LT_4 replacement levels, T3, T4, and TSH levels, T3/T4 ratio, and thyroid autoantibody levels were evaluated both prior and in the third month of the treatment.

Photobiomodulation

A continuous wave GaAIAs-type diode laser [Intelect® Mobile Laser, Model No: 2779, Product Year: 2016; Chattanooga Group (A Division of Encore Medical), Hixton, TN] with a treatment area of 0.07 cm^2 (circular beam shape) was used for all procedures. Output power was 100 mW, wavelength 850 nm, and power density 1.43 W/ cm^2 . The application of laser was made with the patient's neck extended. The boundaries of the thyroid gland were defined by ultrasound and eight target points (superior, mediolateral, inferior border of right and left thyroid poles, right and left side of isthmus) marked with a surgical pen. Stationary applications of low-level laser (28.57 J/ cm^2) were performed over the thyroid gland, which were nearly 1 cm apart from each other. Each application lasted for 20 sec and radiant energy was 2 J, with the tip of the laser held in contact with and perpendicular to the skin. A total of six sessions were applied twice daily for a total of 3 days. The cumulative dose was 96 J (Table 1). Laser was calibrated before each session. Since it was low-level laser, no protection was required.

Statistical analyses

Statistical analyses were performed using SPSS software version 21.0 (Statistical Package for the Social Sciences, Chicago, IL). Continuous variables were expressed as mean $\pm$ standard deviation. Comparison of parametric continuous variables was performed using the Student's *t*-test. The chi-square test was used in the comparison of categorical variables. The correlation coefficients and their significance were calculated using Spearman's test. Logistic regression modeling was used to identify the effective factors over T3/T4 ratio and

TABLE 1. TECHNICAL PARAMETERS OF LASER EQUIPMENT*

Manufacturer	Chattanooga Group (A Division of Encore Medical)
Model identifier	Intelect® Mobile Laser Model 2779
Year produced	2016
Number and type of emitters (laser or LED)	GaAIAs type diode laser
Wavelength and bandwidth, nm	850
Pulse mode, CW or Hz, duty cycle	CW
Beam spot size at target, cm^2	0.012
Treatment area, cm^2	0.07
Irradiance at target, W/ cm^2	1.43
Exposure duration, sec	20
Radiant exposure, J/ cm^2	28.57
Radiant energy, J	2
Number of points irradiated	8
Area irradiated, cm^2	0.56
Application technique	Skin contact
Number and frequency of treatment sessions	6 (twice daily)
Total radiant energy over entire treatment course, J	96

levothyroxine dosage. *p*-Values of 0.05 or less were considered statistically significant.

All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional ethics committee and with the 1964 Helsinki

TABLE 2. PATIENTS' LABORATORY VALUES IN BOTH GROUPS

Total patients (Group 1+Group 2)	
Age, years	37 ± 8
Female/male ratio	340/10
T3, ng/dL	
Before therapy	84.3 ± 8
After therapy	112.6 ± 27
T4, μ g/dL	
Before therapy	9.7 ± 1
After therapy	9.8 ± 1
T3/T4 ratio	
Before therapy	0.009
After therapy	0.012
Vitamin D, ng/dL	
Before therapy	21.6 ± 13
After therapy	61.7 ± 12
Selenium, μ g/L	
Before therapy	69.5 ± 19
After therapy	116.3 ± 21
Iron, μ g/dL	
Before therapy	52.5 ± 28
After therapy	70.9 ± 18
TPO, U/mL	
Before therapy	680.9 ± 561
After therapy	332.2 ± 376
L-Thyroxine dose/week, μ g	
Before therapy	405 ± 258
After therapy	275.9 ± 247

T3, triiodothyronine; T4, thyroxine; TPO, thyroid peroxidase.

Declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all individual participants included in the study.

Results

A total of 210 patients underwent 6 sessions of PBM procedure in Group 1, and 140 patients in Group 2 received dietary supplements only. There were no differences between the two groups in terms of age and gender. Both groups consisted of mostly females. Regardless of the groups, there were improvements in all parameters tested in the study (Table 2). No patient has experienced any side effects.

In Group 1, T3 levels, T3/T4 ratio, vitamin D, selenium, and iron levels increased significantly and TPO Ab and weekly L-thyroxine dosages decreased significantly after treatment (Table 3). In Group 2, T3/T4 ratio and T4 levels did not change significantly (Table 3).

As we compare both groups, the increase in T3 levels and T3/T4 ratio was markedly superior in Group 1 ($p=0.0001$). The decrease in TPO Ab levels was also significantly different between both groups ($p=0.0001$). Hormone replacement needs were also significantly decreased in Group 1 compared with Group 2 ($p=0.03$) (Table 3).

Logistic regression analysis

According to the logistic regression analysis, PBM is 70 times more effective in increasing T3/T4 ratio ($p=0.001$)

and 15 times more effective in decreasing levothyroxine dosage than dietary supplementation alone ($p=0.001$).

Discussion

In our study, the effects of PBM on HT patients were evaluated. As a general result, thyroid functions improved and the patients' needs for LT₄ replacement were reduced.

PBM has been known for a while for its anti-inflammatory properties. Recently, some reports suggest that PBM interacts with the photoacceptors in the oxidative respiratory cycle of living tissues. Following this interaction, ATP production, nitric oxide, and intracellular Ca²⁺ levels increase, stimulating expression of cytokines and growth factors. Overall, cell protection and regeneration occur.⁹⁻¹¹ In a prior study, Parrado et al. showed that PBM stimulates the microcirculation of rat thyroid tissue. Azevedo et al. also demonstrated increased levels of thyroid hormones 7 days after laser application on male mice.^{7,12}

T4 and T3 are both synthesized in the thyroid gland and affect the metabolic rate of most cells in the body. Nearly 80% of the metabolically active form, T3, is converted peripherally from T4 by the enzymatic removal of a single 5' iodine atom (outer ring or 5' monodeiodination).¹³ The T3/T4 ratio is an important indicator of thyroid function. In healthy people without autoimmune thyroiditis, the T3/T4 ratio has been reported as 0.017. In the same study, this rate was significantly lower in patients with autoimmune thyroiditis. This low ratio indicates that there is insufficient conversion from

TABLE 3. CHANGES IN LABORATORY VALUES AFTER TREATMENT IN GROUPS 1 AND 2 AND COMPARISON OF LABORATORY VALUES IN BOTH GROUPS AFTER TREATMENT

	Group 1 (n=210)	p ^a	Group 2 (n=140)	p ^a	p ^b
Age, years	38.4 ± 7		37.9 ± 9		NS
Female/male ratio	206/4		134/6		NS
T3, ng/dL					
Before therapy	84.7 ± 8	0.0001	83.8 ± 3	0.02	NS
After therapy	130.4 ± 21		87.8 ± 12		0.0001
T4, µg/dL					
Before therapy	9.7 ± 7	NS	9.7 ± 1	NS	NS
After therapy	9.8 ± 2		9.3 ± 1		NS
T3/T4 ratio					
Before therapy	0.009	0.0001	0.008	NS	NS
After therapy	0.013		0.008		0.0001
Vitamin D, ng/dL					
Before therapy	20.5 ± 13	0.0001	23 ± 14	0.002	NS
After therapy	62.1 ± 13		61.1 ± 14		NS
Selenium, µg/L					
Before therapy	69 ± 17	0.0001	70.2 ± 17	0.01	NS
After therapy	117.7 ± 20		114.3 ± 21		NS
Iron, µg/dL					
Before therapy	50.7 ± 26	0.0001	55 ± 29	0.0001	NS
After therapy	70.9 ± 18		75 ± 18		NS
TPO, U/mL					
Before therapy	705.9 ± 545	0.0001	645.9 ± 503	0.0001	NS
After therapy	193.8 ± 205		526.1 ± 583		0.0001
L-Thyroxin dose/week, µg					
Before therapy	419.2 ± 255	0.0001	385.1 ± 262	0.0001	NS
After therapy	208.8 ± 220		369.9 ± 253		0.03

^aComparison before and after therapy in both groups.

^bComparison between Group 1 and Group 2.

T4 to T3. The distortion of this transformation can determine the intensity of complaints in Hashimoto thyroiditis.¹⁴ As we approach the normal values of this ratio, there is a positive development in clinical findings of Hashimoto's thyroiditis such as mood and neurophysiological findings. This is the reason why the T3/T4 ratio was evaluated in our study and taken as criteria. In our series, both T3 levels and T3/T4 ratio were increased in Group 1, which were significantly higher than Group 2.

We established a significant reduction in patients' needs for LT₄ replacement therapy, which suggests improved thyroid function and regeneration. All patients demonstrated a decrease in LT₄ doses and 60 (28.5%) patients stopped using LT₄ completely at the end of third month. Höfling et al. also reached a 47% of LT₄ cessation rate in their pilot study.⁴

There was also a reduction of TPO Ab in our series, which is an indicative of decrease in autoimmune response against the thyroid. At the end of third month, there was a decrease in TPO Ab levels in both groups, however, the antibody levels in Group 1 were significantly lower than Group 2, which is attributed to the anti-inflammatory effects of PBM. Some studies in rats showed that laser inhibits production of proinflammatory cytokines such as tumor necrosis factor- α (TNF- α), interleukin (IL)-1 β , IL-2, IL-6, IL-8, and interferon- γ (IFN- γ) by inhibiting gene expression.^{9,15,16} The relationship between IFN- γ and TNF- α producing type 1 T helper cells and high levels of TPO Ab has been known for a while and the suppressive effects of laser on these proinflammatory cytokines may justify the decrease in antibody levels.¹⁷

Conclusions

In conclusion, our results are encouraging and PBM seems to be very effective in increasing T3/T4 ratio and decreasing TPO Ab levels and weekly dosages of LT₄ replacement therapy. Anti-inflammatory properties of PBM are greatly responsible for these changes and PBM causes major improvements in HT-related symptoms of the patient.

Author Disclosure Statement

The authors have no conflicts of interest to declare.

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Address correspondence to:

Candas Ercetin, MD

Department of General Surgery

Bagcilar Training and Research Hospital

University of Health Sciences

Merkez Mah. 6. Sok.

Bagcilar, Istanbul 34200

Turkey

E-mail: ercetin@istanbul.edu.tr

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